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Topic 17: Oscillations

Cambridge International AS & A Level Physics (9702) | Syllabus 2025–2027

Covers: 17.1 Simple harmonic oscillations to 17.3 Damped and forced oscillations, resonance

Level: A Level

Name: _____ Class: _____ Date: _____

How to use this booklet

1. Rate every syllabus statement Red, Amber or Green now, then again after you have worked through the booklet.
2. Draw the diagrams in Section 4 yourself. Drawing and labelling them is a skill the exam tests, so a blank box here is deliberate.
3. Answer the practice questions before you look at the mark scheme. Your teacher has the mark scheme and model diagrams.

Section 1: Syllabus checklist

Every statement in this topic is A Level content. The AS Level topics are assumed knowledge.

#	Syllabus statement (verbatim)	Ref	R	A	G
17.1 Simple harmonic oscillations					
1	understand and use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations, and express the period in terms of both frequency and angular frequency	17.1.1			
2	understand that simple harmonic motion occurs when acceleration is proportional to displacement from a fixed point and in the opposite direction	17.1.2			
3	use $a = -\omega^2 x$ and recall and use, as a solution to this equation, $x = x_0 \sin \omega t$	17.1.3			
4	use the equations $v = v_0 \cos \omega t$ and $v = \pm \omega\sqrt{(x_0^2 - x^2)}$	17.1.4			
5	analyse and interpret graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion	17.1.5			
17.2 Energy in simple harmonic motion					
6	describe the interchange between kinetic and potential energy during simple harmonic motion	17.2.1			
7	recall and use $E = \frac{1}{2}m\omega^2 x_0^2$ for the total energy of a system undergoing simple harmonic motion	17.2.2			
17.3 Damped and forced oscillations, resonance					
8	understand that a resistive force acting on an oscillating system causes damping	17.3.1			
9	understand and use the terms light, critical and heavy damping and sketch displacement–time graphs illustrating these types of damping	17.3.2			
10	understand that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency	17.3.3			

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Section 2: Key definitions

Term	Definition
Displacement, x	Distance from the equilibrium position, in a stated direction. In the equations it is not measured from an arbitrary point
Amplitude, x_0	The maximum displacement from the equilibrium position. It is a positive constant for undamped motion, not another name for displacement
Period, T	Time for one complete oscillation, back to the same point in the motion with the same direction of travel
Frequency, f	Number of oscillations per unit time. f = 1/T. The unit hertz means per second, but the definition is the ratio, not the unit
Angular frequency, ω	$\omega = 2\pi f = 2\pi/T$, in rad s^{-1}. It is not the same quantity as frequency
Phase difference	The fraction of a cycle by which one oscillation leads another, in rad. For $x = x_0 \sin \omega t$ the velocity leads the displacement by π/2 rad, and the acceleration differs in phase from the displacement by π rad
Simple harmonic motion	Motion in which the acceleration is proportional to the displacement from a fixed point and is in the opposite direction: $a = -\omega^2 x$
Total energy of an oscillator	$E = \frac{1}{2}m\omega^2 x_0^2$. For undamped motion it is constant and is shared between kinetic energy and the potential energy of the oscillation
Damping	The decrease of amplitude caused by a resistive force, which removes energy from the oscillating system
Light damping	The system oscillates and the amplitude falls. The motion is still simple harmonic with a decreasing amplitude
Critical damping	The system returns to equilibrium in the shortest time without oscillating
Heavy damping	The system returns to equilibrium without oscillating, more slowly than in critical damping
Natural frequency	The frequency of free oscillation of a system when it is displaced and released
Forced oscillation	Oscillation driven by a periodic external force. The steady frequency is the driving frequency
Resonance	Maximum amplitude of a forced oscillation, occurring when the driving frequency equals the natural frequency

Section 3: Formulae and equations

Quantity	Equation	Symbols and units
Period and frequency (recall)	**$T = 1/f$**	Not printed on the formulae page. T in s, f in Hz
Period and angular frequency (recall)	**$T = 2\pi/\omega$**	Also $\omega = 2\pi f$. Not printed on the formulae page. ω in rad s^{-1}

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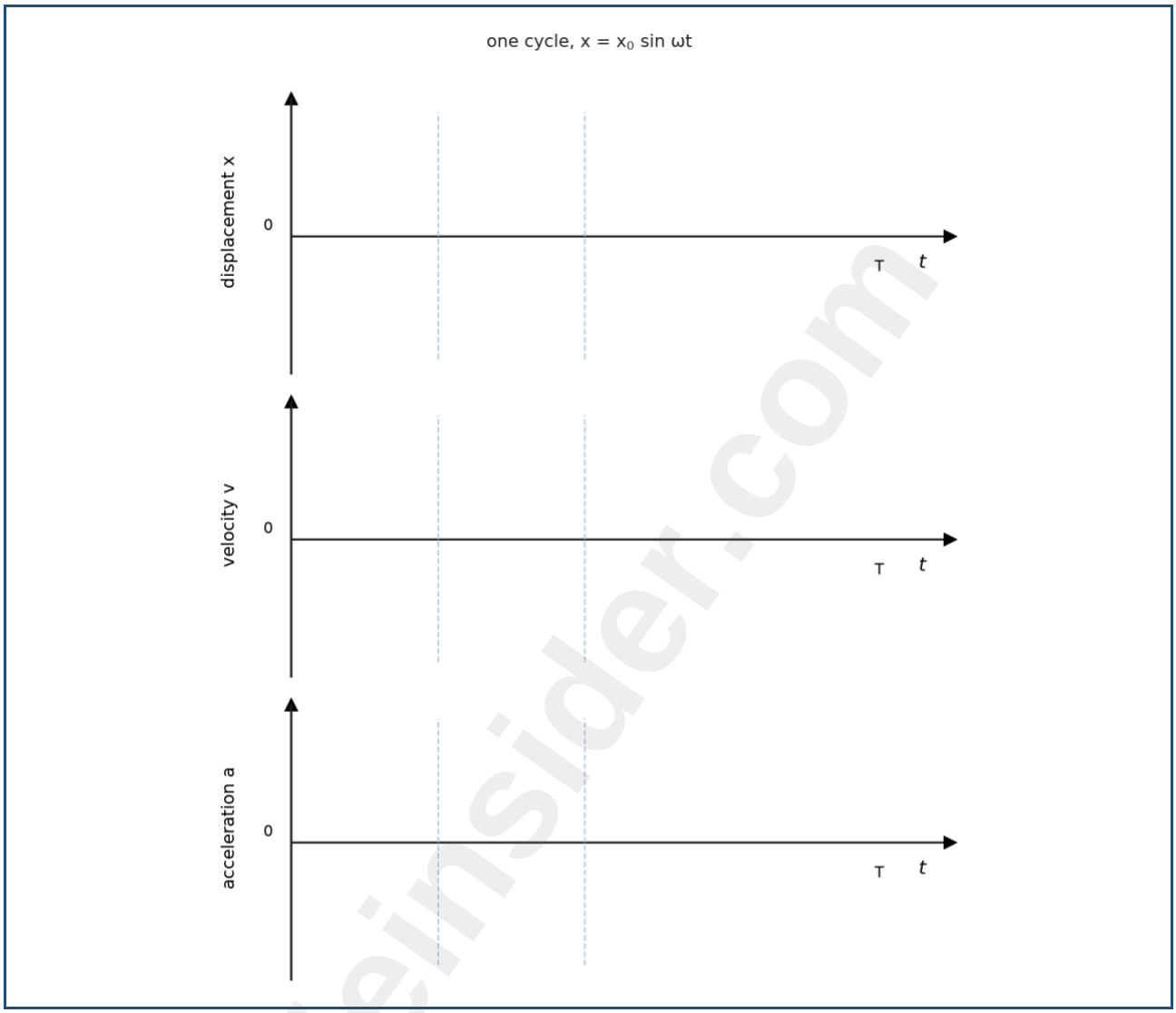
Quantity	Equation	Symbols and units
Defining equation (use)	$\mathbf{a = -\omega^2 x}$	Printed on page 3 of Paper 4, under simple harmonic motion
Displacement solution (recall)	$\mathbf{x = x_0 \sin \omega t}$	x_0 is the amplitude. The outcome says recall and use. The same form is printed on page 3 of Paper 4, but under alternating current/voltage. A cosine solution is the same motion with a different zero of time
Velocity (use)	$\mathbf{v = v_0 \cos \omega t}$	Follows from $x = x_0 \sin \omega t$, with $v_0 = \omega x_0$. Printed on page 3 of Paper 4. The sign is fixed by the choice of t = 0
Velocity at a displacement (use)	$\mathbf{v = \pm\omega\sqrt{(x_0^2 - x^2)}}$	The ± is not chosen by the formula. Speed is the positive value. Printed on page 3 of Paper 4
Maximum acceleration (from $a = -\omega^2 x$)	$\mathbf{a_0 = \omega^2 x_0}$	Magnitude at maximum displacement. Not a separate printed equation
Total energy (recall)	$\mathbf{E = \frac{1}{2} m\omega^2 x_0^2}$	Not printed: recall it. Not the same as $\frac{1}{2}mv^2$. E in J, m in kg, x_0 in m
Kinetic and potential energy (derived)	$\mathbf{E_K = \frac{1}{2} m\omega^2(x_0^2 - x^2)}$, $\mathbf{E_P = \frac{1}{2} m\omega^2 x^2}$	They add to E. E_P here is the potential energy of the oscillation, not gravitational potential energy on its own

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Section 4: Diagrams to draw

4.1 Displacement, velocity and acceleration for one cycle

Task: An object oscillates with $x = x_0 \sin \omega t$. On the three sets of axes, sketch x, v and a against t for one period T. The graphs are in gradient order: the gradient of x is v, and the gradient of v is a. Use that to check the shapes.



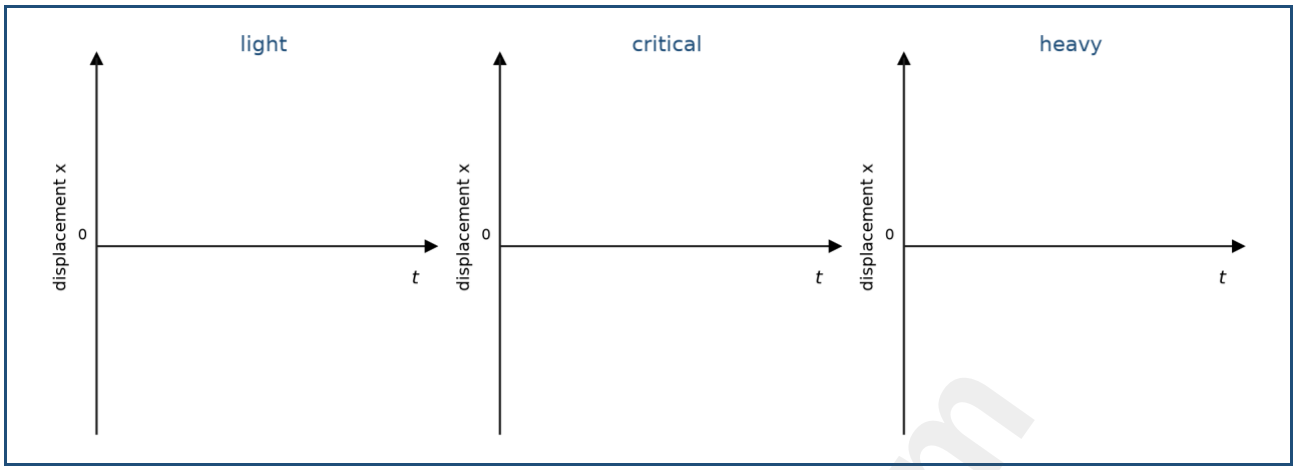
Check your diagram has:

- ☐ x starts at 0, reaches $+x_0$ at T/4, returns to 0 at T/2, reaches $-x_0$ at 3T/4, and is 0 at T
- ☐ v starts at its positive maximum, is 0 at T/4 and at 3T/4, and is negative at T/2
- ☐ a is 0 at t = 0, most negative when x is most positive, and 0 whenever x is 0
- ☐ a is not zero at maximum displacement

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4.2 Light, critical and heavy damping

Task: Each system is released from rest at the same displacement. On the three sets of axes, sketch displacement against time for light, critical and heavy damping.

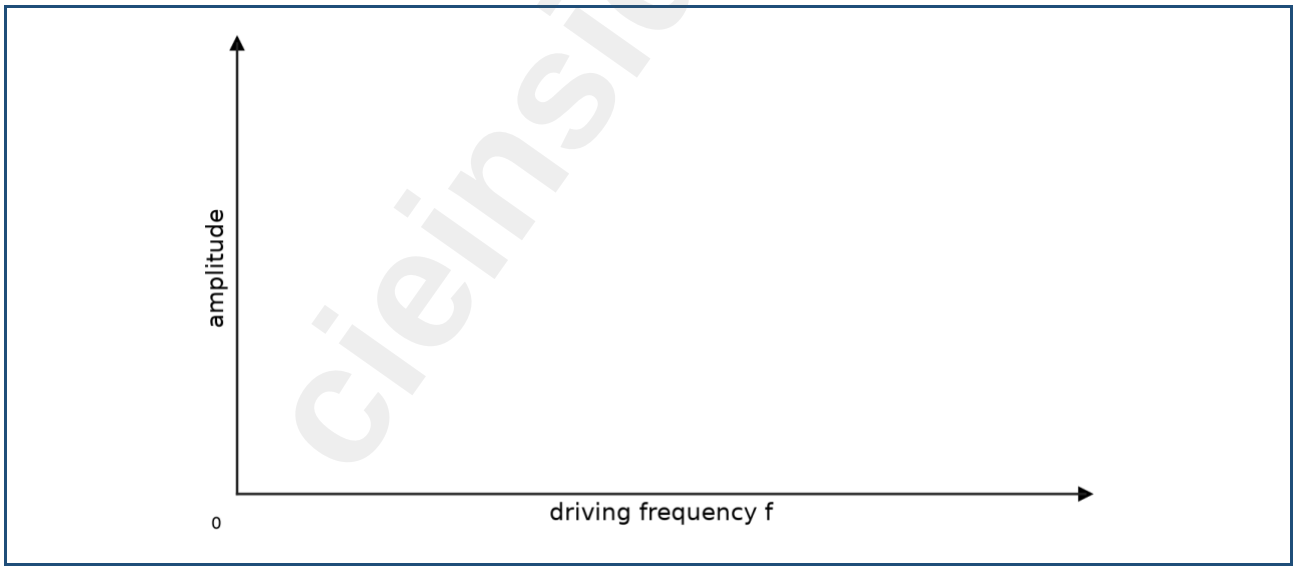


Check your diagram has:

- ☐ Light: the line crosses the time axis repeatedly, with a falling amplitude
- ☐ Critical: the line returns to the time axis without crossing it, and does so sooner than the heavy-damping line
- ☐ Heavy: the line returns to the time axis without crossing it, more slowly than critical damping
- ☐ All three start at the same positive displacement with zero gradient

4.3 Resonance curve

Task: A system is forced to oscillate. Sketch amplitude against driving frequency. Mark the natural frequency on the frequency axis.



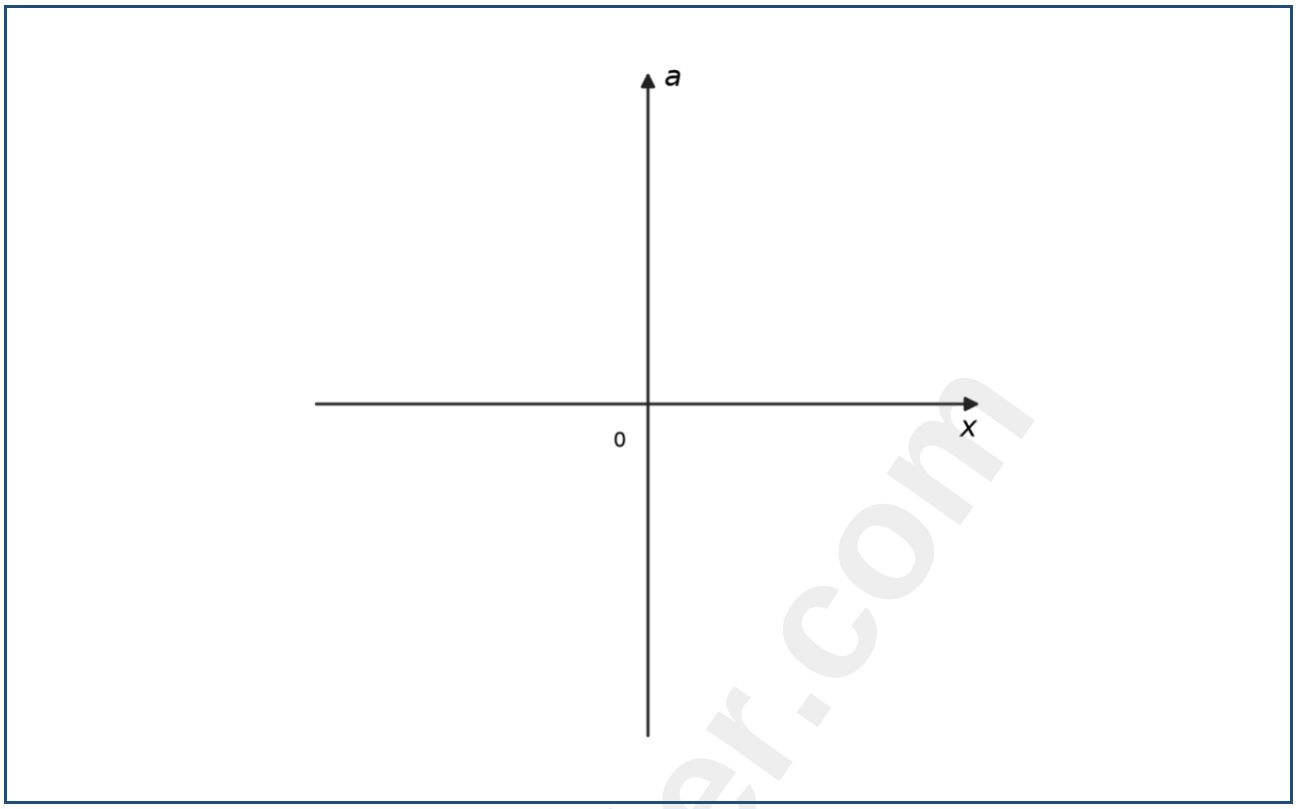
Check your diagram has:

- ☐ Amplitude is not the same at every driving frequency
- ☐ The curve has one maximum
- ☐ The maximum is at the natural frequency
- ☐ The amplitude is smaller both below and above that frequency

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4.4 Acceleration against displacement

Task: Sketch acceleration a against displacement x for simple harmonic motion between the two extreme displacements. Do not continue the line past those points.



Check your diagram has:

- ☐ A single straight line
- ☐ The line passes through the origin
- ☐ Negative gradient: a and x have opposite signs
- ☐ The line stops at $+x_0$ and at $-x_0$
- ☐ Not a closed loop

Section 5: Worked examples

Worked Example 1 A Level

The displacement of a particle is x = 0.030 sin(20t), with x in metres and t in seconds. Find the amplitude, the angular frequency, the period and the frequency. Then find x, v and a at t = 0.040 s.

Compare with $x = x_0 \sin \omega t$. Amplitude $x_0 = 0.030$ m. Angular frequency $\omega = 20$ rad s^{-1}.

$T = 2\pi/\omega = 2\pi/20 = 0.314$ s. $f = 1/T = 20/(2\pi) = 3.18$ Hz.

$v = \omega x_0 \cos \omega t = 0.60 \cos(20t)$. $a = -\omega^2 x = -400x$, so $a = -12 \sin(20t)$. This is $a = -\omega^2 x$.

At t = 0.040 s, $\omega t = 0.80$ rad. sin 0.80 = 0.7174, cos 0.80 = 0.6967.

$x = 0.030 \times 0.7174 = 0.0215$ m. $v = 0.60 \times 0.6967 = 0.418$ m s^{-1}. $a = -12 \times 0.7174 = -8.61$ m s^{-2}.

Check the speed another way: $\omega\sqrt{(x_0^2 - x^2)} = 20 \times \sqrt{(0.030^2 - 0.0215^2)} = 0.418$ m s^{-1}. The sign is positive because cos ωt is positive.

$x_0 = 0.030$ m; $\omega = 20$ rad s^{-1}; $T = 0.314$ s; $f = 3.18$ Hz; at 0.040 s, $x = 0.0215$ m, $v = +0.418$ m s^{-1}, $a = -8.61$ m s^{-2}

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Worked Example 2 A Level

A 0.250 kg mass oscillates with angular frequency 8.0 rad s^{-1} and amplitude 0.050 m. Calculate the total energy, and the kinetic energy and speed when the displacement is 0.030 m.

$E = \frac{1}{2}m\omega^2x_0^2 = \frac{1}{2} \times 0.250 \times 8.0^2 \times 0.050^2 = 0.125 \times 64 \times 0.0025 = 0.0200$ J.

At x = 0.030 m the potential energy of the oscillation is $\frac{1}{2}m\omega^2x^2 = 0.125 \times 64 \times 0.00090 = 0.0072$ J.

Kinetic energy = $E - E_P = 0.0200 - 0.0072 = 0.0128$ J.

Speed from $v = \omega\sqrt{(x_0^2 - x^2)} = 8.0 \times \sqrt{(0.0025 - 0.00090)} = 8.0 \times 0.040 = 0.320$ m s^{-1}.

Check: $\frac{1}{2}mv^2 = \frac{1}{2} \times 0.250 \times 0.320^2 = 0.0128$ J.

E = 0.0200 J; at x = 0.030 m, E_K = 0.0128 J and speed = 0.320 m s^{-1}

Worked Example 3 A Level

An oscillator of mass 0.200 kg has a constant total energy of 0.180 J. The angular frequency changes from 15.0 rad s^{-1} to 10.0 rad s^{-1}. Find the amplitude in each case.

$E = \frac{1}{2}m\omega^2x_0^2$, so $x_0^2 = 2E/(m\omega^2)$.

At 15.0 rad s^{-1}: $\frac{1}{2} \times 0.200 \times 15.0^2 = 22.5$, so $x_0^2 = 0.180/22.5 = 0.00800$ and $x_0 = 0.0894$ m.

At 10.0 rad s^{-1}: $\frac{1}{2} \times 0.200 \times 10.0^2 = 10.0$, so $x_0^2 = 0.180/10.0 = 0.0180$ and $x_0 = 0.134$ m.

The energy did not change, but the amplitude increased because ω decreased. Keeping 0.0894 m would be wrong.

0.0894 m at 15.0 rad s^{-1}; 0.134 m at 10.0 rad s^{-1}

Section 6: Multiple choice

Paper 1 style. Choose one answer, A, B, C or D. 8 marks, one for each question.

1. A body of mass 0.80 kg oscillates with angular frequency 5.0 rad s^{-1} and amplitude 1.50 m. What is the total energy of the oscillation?

A 4.50 J
B 15.0 J
C 22.5 J
D 45.0 J

Answer: _____

2. A particle oscillates with angular frequency 4.0 rad s^{-1} and amplitude 2.0 cm. What is its maximum speed?

A 0.080 m s^{-1}
B 0.16 m s^{-1}
C 0.32 m s^{-1}
D 8.0 m s^{-1}

Answer: _____

3. An object oscillates with frequency 4.00 Hz. What is its angular frequency?

A 0.250 rad s^{-1}
B 4.00 rad s^{-1}
C 12.6 rad s^{-1}
D 25.1 rad s^{-1}

Answer: _____

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4. A particle in simple harmonic motion has angular frequency 4.0 rad s^{-1} and amplitude 0.50 m. What is its speed at a displacement of 0.30 m from equilibrium?

A 0.80 m s^{-1}
B 1.60 m s^{-1}
C 2.00 m s^{-1}
D 2.33 m s^{-1}

Answer: _____

5. An oscillator has angular frequency 10 rad s^{-1} and amplitude 2.5 cm. What is the magnitude of its maximum acceleration?

A 2.50 m s^{-2}
B 5.00 m s^{-2}
C 25.0 m s^{-2}
D 250 m s^{-2}

Answer: _____

6. The displacement of an oscillator is $x = x_0 \sin \omega t$. What is the phase difference between the displacement and the acceleration?

A 0
B $\pi/4$ rad
C $\pi/2$ rad
D π rad

Answer: _____

7. Which statement describes resonance?

A The amplitude is the same at every driving frequency.
B Resonance is the name given to heavy damping.
C The amplitude is maximum when the driving frequency equals the natural frequency.
D The amplitude is maximum when the driving frequency is twice the natural frequency.

Answer: _____

8. An oscillator has angular frequency 20 rad s^{-1}. What is its period?

A 0.0500 s
B 0.314 s
C 3.18 s
D 126 s

Answer: _____

Section 7: Structured questions

Paper 4 style. Total: 62 marks. Show your working in every calculation.

1. Terms used for oscillations.

(a) Define displacement for an oscillation. [1]

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(b) Define amplitude. [1]

(c) Define frequency. [1]

(d) The displacement of an oscillator is $x = x_0 \sin \omega t$. State the phase difference between the displacement and the velocity. [1]

(e) The frequency is 4.00 Hz. Calculate the period and the angular frequency. [2]

2. The acceleration a of a block, in m s^{-2}, is related to its displacement x from equilibrium, in m, by $a = -180x$.

(a) Explain how this equation shows that the motion is simple harmonic. [3]

(b) Determine the angular frequency. [2]

(c) Calculate the frequency. [1]

(d) The amplitude is 12 mm. Calculate the maximum acceleration. [2]

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(e) Calculate the maximum speed. [2]

3. The displacement of a particle is x = 0.050 sin(8.0t), with x in metres and t in seconds.

(a) State the amplitude and the angular frequency. [1]

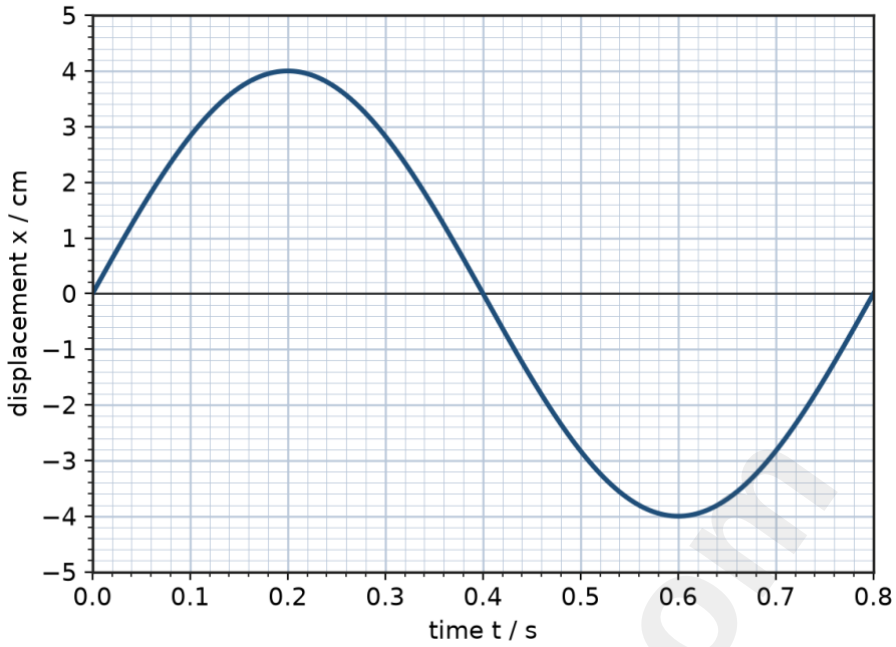
(b) Calculate the displacement at t = 0.10 s. [2]

(c) Calculate the velocity at t = 0.10 s using $v = v_0 \cos \omega t$. [2]

(d) Calculate the velocity at this same displacement using $v = \pm\omega\sqrt{(x_0^2 - x^2)}$. State the sign of your answer and why you chose it. [3]

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4. The graph shows the displacement x of an oscillator against time t.



(a) Determine the amplitude. [1]

(b) Determine the period. [1]

(c) Calculate the angular frequency. [2]

(d) Calculate the maximum speed. [2]

(e) At t = 0.20 s the displacement is at its positive maximum. Calculate the acceleration at this instant. [2]

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5. A mass of 0.150 kg on a vertical spring oscillates with angular frequency 10 rad s^{-1} and amplitude 4.0 cm.

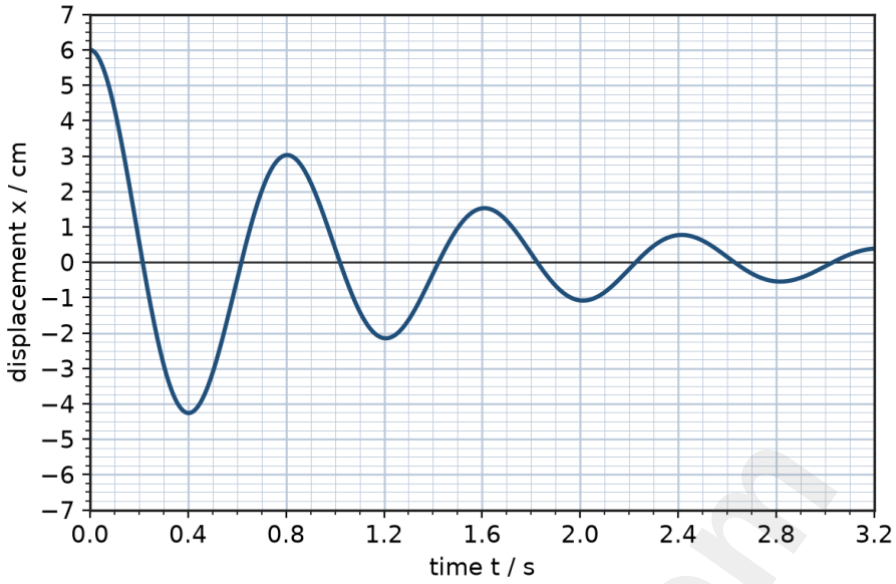
(a) Describe the interchange of kinetic energy and potential energy during one oscillation. State where each is maximum and where each is zero, and state what remains constant. The potential energy is the potential energy of the oscillation. [4]

(b) Calculate the total energy. [2]

(c) Calculate the kinetic energy and the speed at a displacement of 2.0 cm from equilibrium. [3]

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6. The graph shows the displacement of a mass after it is released from rest.



(a) State what causes damping. [1]

(b) Name the type of damping shown on the graph. Give the feature of the graph that identifies it. [2]

(c) State one difference between critical damping and heavy damping. [2]

(d) The resistive force is increased until the motion is heavily damped. State what happens to the energy of the oscillator as it returns to equilibrium. [1]

7. A metal panel has a natural frequency of 28 Hz. It can be driven by a periodic force.

(a) State what is meant by resonance. [2]

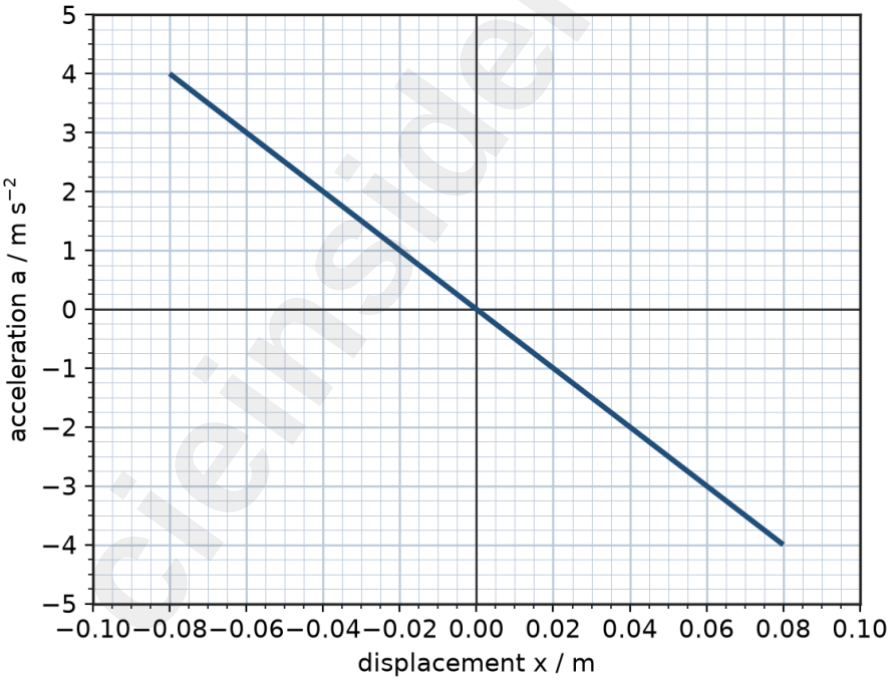
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(b) State the driving frequency at which the amplitude of the panel is maximum. [1]

(c) The panel is driven at 28 Hz and the amplitude stays constant. Explain, in terms of energy, why the driver must continue to supply energy. [2]

(d) The driving frequency is changed to 40 Hz and the driving force has the same amplitude. State how the amplitude of the panel compares with the amplitude at 28 Hz. [1]

8. The graph shows how the acceleration a of a 0.400 kg body depends on its displacement x from equilibrium. The straight line runs from x = −0.080 m to x = +0.080 m.



(a) Explain how the graph shows that the motion is simple harmonic. Refer to the origin and to the gradient. [3]

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(b) Determine the angular frequency. [2]

(c) Determine the maximum speed. [2]

(d) Calculate the total energy of the oscillation. [2]

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Section 8: Examiner tips

Where a tip comes from a Cambridge examiner report for 9702, the exam series and paper are given in brackets.

- In $a = -(\text{constant}) \times x$ the constant is ω^2, not ω. In June 2021, on Paper 41, the most common error was to equate $2k/m$ with ω rather than with ω^2. In June 2024, on Paper 42, a recorded mistake was to include a minus sign inside the square root: take $\omega = \sqrt{(\text{constant})}$, and leave the minus sign in $a = -\omega^2 x$. In March 2022 the significance of the constant values and the minus sign were often omitted. (June 2021, Paper 41; June 2024, Paper 42; March 2022, Paper 42)
- An acceleration–displacement graph shows simple harmonic motion when it is a straight line through the origin with a negative gradient, and it stops at the extreme displacements. In June 2022 many candidates omitted reference to the origin. In June 2023 many closed loops were seen from candidates confusing acceleration with velocity, and many other candidates did not end the line at the maximum values. (June 2022, Paper 41; June 2023, Paper 42)
- $E = \frac{1}{2}m\omega^2 x_0^2$. If the total energy stays the same while ω decreases, the amplitude increases. In June 2022, on Paper 41, others simply said the amplitude would remain the same because the energy remained the same. (June 2022, Paper 41)
- Recall $E = \frac{1}{2}m\omega^2 x_0^2$. It is not one of the three simple-harmonic equations grouped with $a = -\omega^2 x$. In Nov 2022 many candidates were unable to recall the equation for the total energy. Where it was used, mistakes included not squaring the amplitude and forgetting to convert the amplitude into metres. (Nov 2022, Paper 42)
- For a mass on a vertical spring, the potential energy of the oscillation is elastic and gravitational together. It is maximum at both ends and minimum at equilibrium, not maximum at the top and minimum at the bottom. In June 2023 that was a common misconception, and the terms amplitude and displacement were commonly conflated. Amplitude is the maximum displacement from equilibrium. (June 2023, Paper 42)
- A resistive force causes damping by removing energy, and that is why the amplitude falls. In June 2024 few candidates described that energy reduction, and a significant number did not name the type of damping. Light damping still oscillates. In Nov 2021 the motion was described as simple harmonic before the damping began and then not simple harmonic after. Critical damping returns to equilibrium in the shortest time without oscillating. Heavy damping returns more slowly, also without oscillating. In Nov 2023 some graphs drawn for critical damping were undamped, lightly damped or overdamped, and in Nov 2025 many candidates drew light damping when heavy damping was required. (June 2024, Paper 41; Nov 2021, Paper 43; Nov 2023, Paper 43; Nov 2025, Paper 42)
- Resonance needs both a maximum amplitude and a driving frequency equal to the natural frequency. In June 2024 fewer candidates explained that the oscillations reach maximum amplitude. Resonance is not damping: in June 2022 there was some confusion between resonance and damping. The amplitude is not the same at every driving frequency. In June 2021 a significant number of candidates showed that misconception. (June 2024, Paper 41; June 2022, Paper 42; June 2021, Paper 41)
- Frequency and angular frequency are different quantities. $T = 1/f$ and $T = 2\pi/\omega$, so $\omega = 2\pi f$. In June 2025 weaker candidates often confused the terms frequency and angular frequency. Define frequency as the number of oscillations per unit time. In Nov 2025 some candidates specified 'per second' and did not gain credit. (June 2025, Paper 44; Nov 2025, Paper 44)
- In $a = -\omega^2 x$ and in $x = x_0 \sin \omega t$, x is the displacement from the equilibrium position. In Nov 2022 some candidates described x only as displacement. On a displacement–time sketch, label the negative half of the axis as well as the positive half, and take the period as one complete cycle. In June 2022 the calculated period was treated as the time for half a cycle, and only the positive half of the displacement axis was labelled. (Nov 2022, Paper 41; June 2022, Paper 42)

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